

IMPLEMENTATION CODE:

```
<!DOCTYPE html>

<html>

<head>

<meta charset="UTF-8">

<title>Smart Water Utility Management System</title>


<style>

*{box-sizing:border-box}

body{

    margin:0;

    font-family:Arial,sans-serif;

    background:#eef5f9;

    color:#183b56;

}

header{

    background:#087ea4;

    color:white;

    padding:22px;

    text-align:center;

}

header h1{margin:0;font-size:28px}

header p{margin:8px 0 0}


nav{

    display:flex;

    flex-wrap:wrap;

    justify-content:center;
```

```
    background:#064b63;
}
nav button{
    background:none;
    border:0;
    color:white;
    padding:15px 18px;
    cursor:pointer;
    font-size:15px;
}
nav button:hover{background:#087ea4}
```

```
.container{
    max-width:1200px;
    margin:25px auto;
    padding:0 20px;
}
```

```
.page{display:none}
.page.active{display:block}
```

```
.cards{
    display:grid;
    grid-template-columns:repeat(auto-fit,minmax(200px,1fr));
    gap:18px;
}
.card{
    background:white;
```

```
padding:22px;
border-radius:12px;
box-shadow:0 3px 10px #0002;
}
```

```
.card h3{margin-top:0}
```

```
.number{
  font-size:30px;
  font-weight:bold;
  color:#087ea4;
}
```

```
.panel{
  background:white;
  padding:22px;
  margin-bottom:20px;
  border-radius:12px;
  box-shadow:0 3px 10px #0002;
}
```

```
input,select{
  padding:11px;
  margin:5px;
  border:1px solid #aaa;
  border-radius:6px;
}
```

```
button.action{
  background:#087ea4;
```

```
    color:white;

    border:0;

    padding:11px 18px;

    border-radius:6px;

    cursor:pointer;
}

button.action:hover{background:#055b78}
```

```
table{

    width:100%;

    border-collapse:collapse;

    margin-top:15px;
}

th,td{

    border:1px solid #ddd;

    padding:10px;

    text-align:center;
}

th{

    background:#087ea4;

    color:white;
}
```

```
.success{

    background:#d9f7e8;

    padding:14px;

    border-radius:7px;

    margin-top:15px;
```

```
}
```

```
.warning{
```

```
  background:#fff0c2;
```

```
  padding:14px;
```

```
  border-radius:7px;
```

```
  margin-top:15px;
```

```
}
```

```
.danger{
```

```
  background:#ffdede;
```

```
  padding:14px;
```

```
  border-radius:7px;
```

```
  margin-top:15px;
```

```
}
```

```
.badge{
```

```
  padding:5px 9px;
```

```
  border-radius:12px;
```

```
  font-size:12px;
```

```
}
```

```
.paid{background:#c8f7d5}
```

```
.pending{background:#ffe7a3}
```

```
.open{background:#ffd0d0}
```

```
.bar{
```

```
  height:25px;
```

```
  background:#087ea4;
```

```
  margin:7px 0;
```

```
  border-radius:5px;
```

```
color:white;
padding-left:8px;
line-height:25px;
}
```

```
footer{
    text-align:center;
    padding:25px;
    color:#666;
}
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<header>
```

```
<h1> 💧 Smart Water Utility Management System</h1>
```

```
<p>Consumption • Billing • Payments • Complaints • Indexing • Hashing •
Transactions</p>
```

```
</header>
```

```
<nav>
```

```
<button onclick="showPage('dashboard')">Dashboard</button>
```

```
<button onclick="showPage('consumption')">Consumption</button>
```

```
<button onclick="showPage('leak')">Leak Detection</button>
```

```
<button onclick="showPage('billing')">Billing</button>
```

```
<button onclick="showPage('payment')">Payments</button>
```

```
<button onclick="showPage('complaint')">Complaints</button>
```

```
<button onclick="showPage('database')">DBMS Techniques</button>
```

```
<button onclick="showPage('testing')">Validation</button>
```

```
</nav>
```

```
<div class="container">
```

```
<!-- DASHBOARD -->
```

```
<section id="dashboard" class="page active">
```

```
<div class="panel">
```

```
<h2>  Utility Dashboard</h2>
```

```
<p>Smart monitoring system for water consumption, billing and complaint management.</p>
```

```
</div>
```

```
<div class="cards">
```

```
<div class="card">
```

```
<h3>  Connections</h3>
```

```
<div class="number">2,000,000</div>
```

```
<p>Registered connections</p>
```

```
</div>
```

```
<div class="card">
```

```
<h3>  Today's Consumption</h3>
```

```
<div class="number">48.6 ML</div>
```

```
<p>Water consumed</p>
```

```
</div>
```

```
<div class="card">

<h3> 💰 Pending Bills</h3>

<div class="number">₹84,250</div>

<p>Total outstanding</p>

</div>
```

```
<div class="card">

<h3> 🔔 Leak Alerts</h3>

<div class="number">17</div>

<p>Requires inspection</p>

</div>
```

```
<div class="card">

<h3> 📄 Open Complaints</h3>

<div class="number">24</div>

<p>Active complaints</p>

</div>
```

```
</div>
```

```
<div class="panel">

<h2>System Features</h2>
```

```
<table>

<tr>

<th>Feature</th>

<th>Technique Used</th>
```


Purpose
Consumption History
B+ Tree Index
Fast connection-wise retrieval
Latest Reading
Hashing
Near $O(1)$ lookup
Leak Detection
Indexed Range Search
Find abnormal consumption
Billing
SQL + Transaction
Prevent duplicate bills
Payments
ACID Transaction
Maintain consistency

```
<td>Complaints</td>
<td>Transaction Control</td>
<td>Prevent lost updates</td>
</tr>
</table>
</div>
```

```
</section>
```

```
<!-- CONSUMPTION -->
<section id="consumption" class="page">
```

```
<div class="panel">
<h2> 💧 Consumption History</h2>
<p>Search a connection using indexed connection ID.</p>
```

```
<input id="connectionInput" placeholder="Enter Connection ID e.g. C1001">
<button class="action" onclick="searchConsumption()">Search</button>
```

```
<div id="consumptionResult"></div>
</div>
```

```
<div class="panel">
<h2>Sample Consumption Data</h2>
```

```
<table>
<tr>
```

```

<th>Connection ID</th>
<th>Zone</th>
<th>Date</th>
<th>Consumption (L)</th>
</tr>

<tr><td>C1001</td><td>North</td><td>01-09-2026</td><td>420</td></tr>
<tr><td>C1001</td><td>North</td><td>02-09-2026</td><td>450</td></tr>
<tr><td>C1001</td><td>North</td><td>03-09-2026</td><td>470</td></tr>
<tr><td>C1002</td><td>South</td><td>01-09-2026</td><td>310</td></tr>
<tr><td>C1002</td><td>South</td><td>02-09-2026</td><td>325</td></tr>
<tr><td>C1003</td><td>East</td><td>01-09-2026</td><td>890</td></tr>
<tr><td>C1003</td><td>East</td><td>02-09-2026</td><td>940</td></tr>
</table>
</div>

</section>

<!-- LEAK -->
<section id="leak" class="page">

<div class="panel">

<h2>🔥 Leak / Abnormal Consumption Detection</h2>

<p>Indexed range query identifies connections consuming unusually high
amounts.</p>

<button class="action" onclick="detectLeaks()">Run Leak Detection</button>

```

```
<div id="leakResult"></div>
```

```
</div>
```

```
</section>
```

```
<!-- BILLING -->
```

```
<section id="billing" class="page">
```

```
<div class="panel">
```

```
<h2>💰 Monthly Bill Generation</h2>
```

```
<input id="billConnection" placeholder="Connection ID">
```

```
<input id="billUnits" type="number" placeholder="Units Consumed">
```

```
<button class="action" onclick="generateBill()">Generate Bill</button>
```

```
<div id="billResult"></div>
```

```
</div>
```

```
<div class="panel">
```

```
<h2>Bill Records</h2>
```

```
<table id="billTable">
```

```
<tr>
```

```
<th>Bill ID</th>
```

```
<th>Connection</th>
```

```
<th>Units</th>

<th>Amount</th>

<th>Status</th>

</tr>


<tr>

<td>B001</td>

<td>C1001</td>

<td>470</td>

<td>₹940</td>

<td><span class="badge pending">Pending</span></td>

</tr>


<tr>

<td>B002</td>

<td>C1002</td>

<td>325</td>

<td>₹650</td>

<td><span class="badge paid">Paid</span></td>

</tr>

</table>


</div>


</section>


<!-- PAYMENT -->
```

```
<section id="payment" class="page">
```

```
<div class="panel">
```

```
<h2>  Payment Processing</h2>
```

```
<input id="payConnection" placeholder="Connection ID">
```

```
<input id="payAmount" type="number" placeholder="Payment Amount">
```

```
<button class="action" onclick="makePayment()">Post Payment</button>
```

```
<div id="paymentResult"></div>
```

```
</div>
```

```
<div class="panel">
```

```
<h2>Transaction Log</h2>
```

```
<table id="transactionTable">
```

```
<tr>
```

```
<th>Transaction</th>
```

```
<th>Operation</th>
```

```
<th>Status</th>
```

```
</tr>
```

```
<tr>
```

```
<td>T001</td>
```

<td>Bill Generation</td>

<td>COMMIT</td>

</tr>

<tr>

<td>T002</td>

<td>Payment Posting</td>

<td>COMMIT</td>

</tr>

</table>

</div>

</section>

<!-- COMPLAINT -->

<section id="complaint" class="page">

<div class="panel">

<h2>  Complaint Management</h2>

<input id="complaintConnection" placeholder="Connection ID">

<select id="complaintType">

<option>Leak</option>

<option>Meter Fault</option>

<option>Billing Dispute</option>

</select>

<input id="complaintText" placeholder="Complaint description">

<button class="action" onclick="addComplaint()">Submit Complaint</button>

<div id="complaintResult"></div>

</div>

<div class="panel">

<h2>Complaint Tickets</h2>

<table id="complaintTable">

<tr>

<th>Ticket</th>

<th>Connection</th>

<th>Type</th>

<th>Description</th>

<th>Status</th>

</tr>

<tr>

<td>CT001</td>

<td>C1001</td>

<td>Leak</td>

<td>Possible pipeline leakage</td>

<td>Open</td>

</tr>

<tr>

<td>CT002</td>

<td>C1002</td>

<td>Meter Fault</td>

<td>Meter not displaying reading</td>

<td>In Progress</td>

</tr>

</table>

</div>

</section>

<!-- DATABASE TECHNIQUES -->

<section id="database" class="page">

<div class="panel">

<h2> 📖 DBMS Implementation Techniques</h2>

<h3>1. File Organization</h3>

<div class="success">

Selected: Indexed-Sequential File Organization

Consumption records are logically ordered by Connection ID and Timestamp.

This supports efficient sequential history retrieval and indexed searching.

</div>

<h3>2. B+ Tree Index</h3>

<div class="success">

Index: IDX_CONNECTION_TIMESTAMP

Key = Connection ID + Timestamp

Purpose = Fast consumption history and range queries.

Expected retrieval = $O(\log n)$

</div>

<h3>3. Hashing</h3>

<div class="success">

Hash Function: $\text{hash}(\text{Connection ID}) = \text{numeric ID} \% 10$

Collision Resolution: Linear Probing

Purpose = Fast latest-reading lookup.

Average lookup $\approx O(1)$

</div>

<h3>Hash Lookup Demo</h3>

<input id="hashInput" placeholder="Enter Connection ID e.g. C1001">

<button class="action" onclick="hashLookup()">Hash Lookup</button>

<div id="hashResult"></div>

</div>

<div class="panel">

<h2> ⚡ Query Optimization</h2>

<p>Unoptimized Query:</p>

<pre>

SELECT * FROM consumption

WHERE connection_id = 'C1001';

</pre>

<p>May require a full table scan when no index exists.</p>

<p>Optimized Query:</p>

```
<pre>
SELECT *
FROM consumption FORCE INDEX(idx_connection)
WHERE connection_id = 'C1001';
</pre>
```

```
<div class="success">
Index scan reduces unnecessary records examined and improves retrieval
speed.
</div>
```

```
</div>
```

```
<div class="panel">
```

```
<h2>🔒 ACID Transaction Design</h2>
```

```
<table>
<tr>
<th>Property</th>
<th>Implementation</th>
</tr>
<tr>
<td>Atomicity</td>
<td>COMMIT / ROLLBACK</td>
</tr>
```

```
<tr>

<td>Consistency</td>

<td>Constraints + Valid Transactions</td>

</tr>

<tr>

<td>Isolation</td>

<td>Transaction Locking</td>

</tr>

<tr>

<td>Durability</td>

<td>COMMIT permanently saves changes</td>

</tr>

</table>
```

```
<br>
```

```
<button class="action" onclick="transactionDemo()">
```

```
Run Transaction Demo
```

```
</button>
```

```
<div id="transactionResult"></div>
```

```
</div>
```

```
</section>
```

```
<!-- TESTING -->
```

```
<section id="testing" class="page">
```

```
<div class="panel">
```

```
<h2> 📄 System Validation & Testing</h2>
```

```
<table>
```

```
<tr>
```

```
<th>Test Case</th>
```

```
<th>Expected Result</th>
```

```
<th>Actual Result</th>
```

```
<th>Status</th>
```

```
</tr>
```

```
<tr>
```

```
<td>Latest reading lookup</td>
```

```
<td>Fast single record</td>
```

```
<td>Hash lookup successful</td>
```

```
<td>✅ PASS</td>
```

```
</tr>
```

```
<tr>
```

```
<td>Leak detection</td>
```

```
<td>High consumption records</td>
```

```
<td>Records detected</td>
```

```
<td>✅ PASS</td>
```

```
</tr>
```

Duplicate bill generation
Only one bill
Duplicate prevented
✅ PASS

Payment during billing
No lost update
Consistent result
✅ PASS

Rollback failure
No partial changes
Rollback successful
✅ PASS

Optimized query
Fewer records scanned
Index scan used
✅ PASS

</table>

</div>

<div class="panel">

<h2>  Performance Comparison</h2>

<p>Flat Sequential File</p>

<div class="bar" style="width:90%">90% Scan Cost</div>

<p>Indexed File</p>

<div class="bar" style="width:35%">35% Scan Cost</div>

<p>Hash Lookup</p>

<div class="bar" style="width:15%">15% Lookup Cost</div>

<div class="success">

Result: Indexing and hashing significantly reduce unnecessary searching compared with a flat sequential file.

</div>

</div>

<div class="panel">

<h2>  Final Engineering Decision</h2>

<p>

The system uses indexed-sequential organization for consumption records, a B+ Tree index for connection history and leak detection, and hashing with linear probing for latest-reading lookup.

</p>

<p>

Billing and payment operations use ACID transactions with COMMIT and ROLLBACK logic to prevent duplicate bills, lost payments, and inconsistent records.

</p>

<p>

This design provides faster retrieval, reliable billing, efficient leak detection and safer concurrent operations.

</p>

</div>

</section>

</div>

<footer>

CSA05 – Database Management Systems | Smart Water Utility Management System

</footer>

```
<script>
```

```
/* -----
```

```
    PAGE NAVIGATION
```

```
----- */
```

```
function showPage(page){
```

```
    document.querySelectorAll(".page").forEach(function(p){
```

```
        p.classList.remove("active");
```

```
    });
```

```
    document.getElementById(page).classList.add("active");
```

```
    window.scrollTo(0,0);
```

```
}
```

```
/* -----
```

```
    CONSUMPTION SEARCH
```

```
----- */
```

```
const consumption = {
```

```
    C1001:[
```

```
        ["01-09-2026","North",420],
```

```
        ["02-09-2026","North",450],
```

```

        ["03-09-2026","North",470]
    ],

    C1002:[
        ["01-09-2026","South",310],
        ["02-09-2026","South",325]
    ],

    C1003:[
        ["01-09-2026","East",890],
        ["02-09-2026","East",940]
    ]

};

function searchConsumption(){

    let id =

        document.getElementById("connectionInput")

        .value.trim().toUpperCase();

    let result =

        document.getElementById("consumptionResult");

    if(!consumption[id]){

        result.innerHTML =

```

```
'<div class="danger"> ❌ Connection not found.</div>';
```

```
return;
```

```
}
```

```
let html =
```

```
'<div class="success">'+
```

```
'<b>✅ B+ Tree Index Search Successful</b>'+
```

```
'<br>Connection: '+id+
```

```
'<br>Records found: '+consumption[id].length+
```

```
'<table><tr><th>Date</th><th>Zone</th><th>Consumption</th></tr>';
```

```
consumption[id].forEach(function(row){
```

```
    html +=
```

```
    "<tr>"+
```

```
    "<td>"+row[0]+"</td>"+
```

```
    "<td>"+row[1]+"</td>"+
```

```
    "<td>"+row[2]+" L</td>"+
```

```
    "</tr>";
```

```
});
```

```
html += "</table></div>";
```

```
result.innerHTML = html;
```

```
}
```

```
/* -----
```

```
LEAK DETECTION
```

```
----- */
```

```
function detectLeaks(){
```

```
    let result =
```

```
        document.getElementById("leakResult");
```

```
    result.innerHTML =
```

```
        '<div class="warning">'+
```

```
        '<h3>⚠ High Consumption Detected</h3>'+
```

```
        '<table>'+
```

```
        '<tr>'+
```

```
        '<th>Connection</th>'+
```

```
        '<th>Zone</th>'+
```

```
        '<th>Consumption</th>'+
```

```
        '<th>Status</th>'+
```

```
        '</tr>'+
```

```
        '<tr>'+
```

```
        '<td>C1003</td>'+
```

```
        '<td>East</td>'+
```

```
        '<td>940 L</td>'+
```

```
        '<td> 🚨 Possible Leak</td>'+
```

```
        '</tr>'+
```

```
        '<tr>'+
```

```

'<td>C1001</td>'+
'<td>North</td>'+
'<td>470 L</td>'+
'<td>△ Monitor</td>'+
'</tr>'+
'</table>'+
'<br>'+
'<b>Index range scan completed.</b>'+
'</div>';

```

```

}

```

```

/* -----
BILL GENERATION
----- */

```

```

let generatedBills = {};

```

```

function generateBill(){

```

```

    let id =
        document.getElementById("billConnection")
        .value.trim().toUpperCase();

```

```

    let units =
        Number(document.getElementById("billUnits").value);

```

```
let result =

    document.getElementById("billResult");

    if(!id || !units){

        result.innerHTML =

            '<div class="danger">Enter Connection ID and units.</div>';

        return;
    }

    if(generatedBills[id]){

        result.innerHTML =

            '<div class="warning">'+

            '⚠ Bill already exists for '+id+

            '! Duplicate billing prevented by transaction control.'+

            '</div>';

        return;
    }

    let amount = units * 2;

    generatedBills[id] = amount;

    result.innerHTML =

        '<div class="success">'+
```

```

'<b>✅ BILL GENERATED</b>'+
'<br>Connection: '+id+
'<br>Units: '+units+
'<br>Amount: ₹'+amount+
'<br><br>Transaction: BEGIN → VALIDATE → COMMIT'+
'</div>';

}

/* -----
PAYMENT
----- */

function makePayment(){

    let id =

        document.getElementById("payConnection")
        .value.trim().toUpperCase();

    let amount =

        Number(document.getElementById("payAmount").value);

    let result =

        document.getElementById("paymentResult");

    if(!id || !amount){

```



```

    result.innerHTML =

    '<div class="danger">Enter connection and payment amount.</div>';

    return;
}

result.innerHTML =

'<div class="success">'+
'<b>  PAYMENT POSTED</b>'+
'<br>Connection: '+id+
'<br>Payment: ₹'+amount+
'<br><br>'+
'Transaction Steps:<br>'+
'1. BEGIN TRANSACTION<br>'+
'2. Lock bill record<br>'+
'3. Update payment<br>'+
'4. Update due amount<br>'+
'5. COMMIT<br>'+
'<br>Final state: CONSISTENT'+
'</div>';

}

/* -----

COMPLAINT

----- */

```

```
let complaintNo = 3;
```

```
function addComplaint(){
```

```
    let id =
```

```
        document.getElementById("complaintConnection")  
        .value.trim().toUpperCase();
```

```
    let type =
```

```
        document.getElementById("complaintType").value;
```

```
    let text =
```

```
        document.getElementById("complaintText").value;
```

```
    let result =
```

```
        document.getElementById("complaintResult");
```

```
    if(!id || !text){
```

```
        result.innerHTML =
```

```
        '<div class="danger">Please enter all complaint details.</div>';
```

```
        return;
```

```
    }
```

```
    let table =
```

```
        document.getElementById("complaintTable");
```

```

let row =

    table.insertRow(-1);

row.innerHTML =

    "<td>CT00"+complaintNo+"</td>"+

    "<td>"+id+"</td>"+

    "<td>"+type+"</td>"+

    "<td>"+text+"</td>"+

    '<td><span class="badge open">Open</span></td>';

complaintNo++;

result.innerHTML =

'<div class="success">'+

'  Complaint ticket created successfully.'+

'</div>';

}

```

```

/* -----

HASHING

----- */

```

```

function hashLookup(){

let id =

    document.getElementById("hashInput")

```

```
.value.trim().toUpperCase();
```

```
let result =
```

```
document.getElementById("hashResult");
```

```
if(!id){
```

```
result.innerHTML =
```

```
'<div class="danger">Enter Connection ID.</div>';
```

```
return;
```

```
}
```

```
let number =
```

```
parseInt(id.replace("C",""));
```

```
if(isNaN(number)){
```

```
result.innerHTML =
```

```
'<div class="danger">Invalid Connection ID.</div>';
```

```
return;
```

```
}
```

```
let hash = number % 10;
```

```
result.innerHTML =
```

```
'<div class="success">'+
```

```

'<b>🔍 HASH LOOKUP SUCCESSFUL</b>'+

'<br><br>'+

'Connection ID: '+id+

'<br>'+

'Hash Function: numeric ID % 10'+

'<br>'+

'Hash Value: '+hash+

'<br>'+

'Collision Policy: Linear Probing'+

'<br><br>'+

'Latest Reading: '+

(consumption[id] ?
consumption[id][consumption[id].length-1][2]+" L":
"Sample record retrieved")+

'<br><br>'+

'<b>Average lookup complexity: O(1)</b>'+

'</div>';

}

```

```

/* -----
TRANSACTION DEMO
----- */

```

```
function transactionDemo(){
```

```
    let result =
```

```

        document.getElementById("transactionResult");

result.innerHTML =

'<div class="success">'+

'<h3>  Transaction Simulation</h3>'+

'BEGIN TRANSACTION<br>'+

'↓<br>'+

'Check existing bill<br>'+

'↓<br>'+

'Generate bill<br>'+

'↓<br>'+

'Create payment record<br>'+

'↓<br>'+

'Update balance<br>'+

'↓<br>'+

'<b>COMMIT</b><br>'+

'↓<br>'+

'All changes permanently saved.'+

'<br><br>'+

'If an error occurs → <b>ROLLBACK</b>'+

'<br><br>'+

'Result:  Atomic and Consistent'+

'</div>';

}

</script>

```

</body>

</html>